Network Manager Release Notes

Version 2.6

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1: Introduction

This document provides a summary of the major changes in this release of Network Manager for Telecoms.

1.1 Intended Audience

This document is intended for Administrators and Developers with a detailed technical knowledge of IQGeo Platform and utilities.

2: Installation

The following changes relate to installation and upgrade.

2.1 Prerequisites

Component	Version	Comments
IQGeo Platform	6.5	Mandatory
IQGeo Workflow Module	6.5	Mandatory
IQGeo Anywhere	6.5 (or higher)	Optional

All components must include any released patches.

2.2 Installing the Software

1. Uncompress the software download and copy the **comms** folder into:

   ```
   <myWorld_install_dir>/WebApps/myworldapp/modules
   ```

2. Download the module's JavaScript dependencies:

   ```
   myw_product fetch node_modules
   ```

3. Rebuild your JavaScript bundles:

   ```
   myw_product build applications
   ```

2.3 Creating a New Database

If you do not already have an IQGeo database, or that database does not yet have the Network Manager tables installed, you will need to:

- Install the Network Manager tables.
- Configure the network object types for your installation.

These processes are covered in detail in the *Installation and Configuration Guide*.

2.4 Upgrading an Existing Database

If you already have a Network Manager database from a previous release, you will need to:

- Upgrade your database.
- Perhaps perform some post-upgrade configuration changes.

These processes are covered in detail in *Upgrading* on page 28.

3: End-user GUI Changes

The following changes have been made to the end-user GUI.

3.1 Adding an Object

Features in the **Add object** dialog are now grouped by category (structures, cables and so on). Equipment features are available if a compatible housing (structure) is the current feature.

Click the **Add Object** icon on the Network Manager toolbar.

The **Add Object** dialog appears, presenting an overview of the available object types grouped per type.

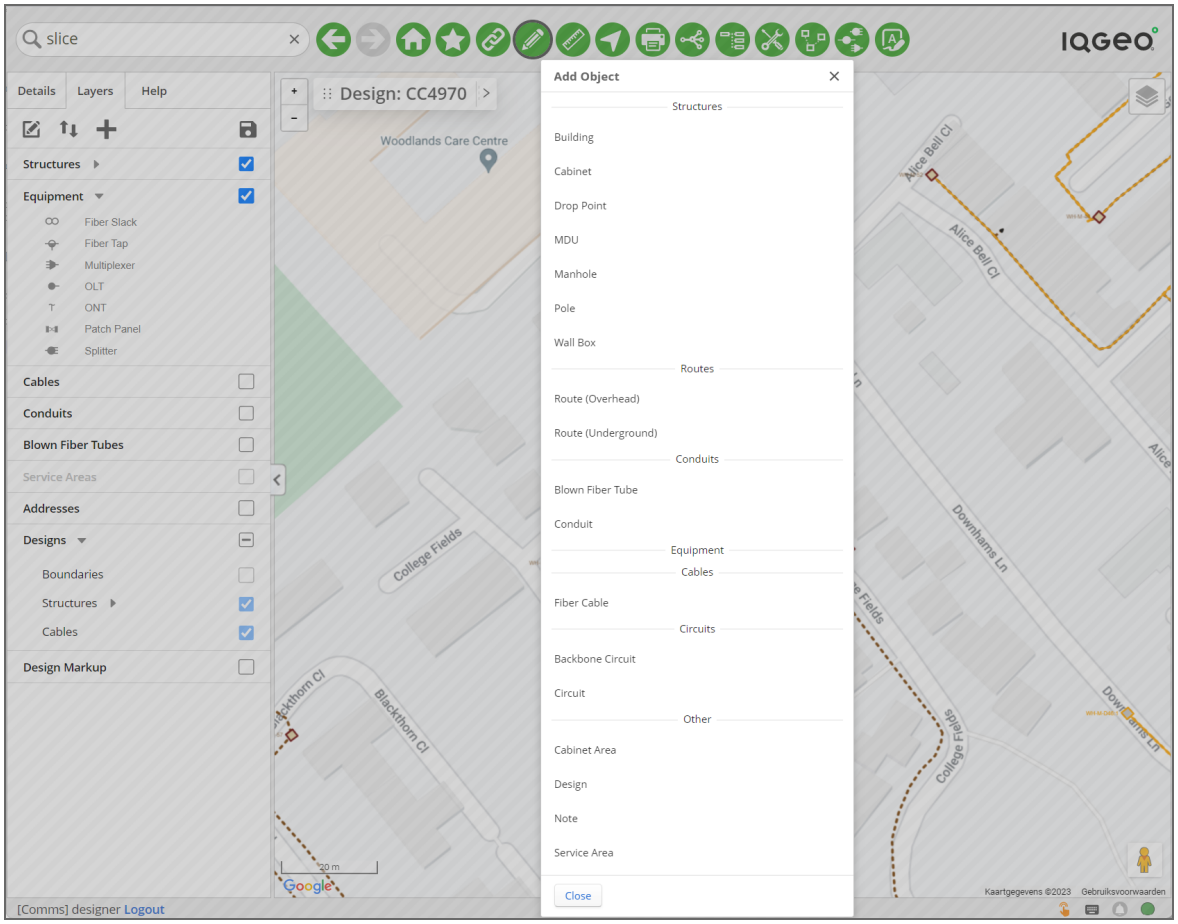

Figure 3-1: Add Object Dialog

From that list, select the object type that you want to add.

> Regarding **Equipment**, it is important to know that this section of the list is only populated, if a compatible type of containing structure is selected.

The three examples below, illustrate respectively, a situation with *no structure* selected, with a manhole selected and with a pole selected.

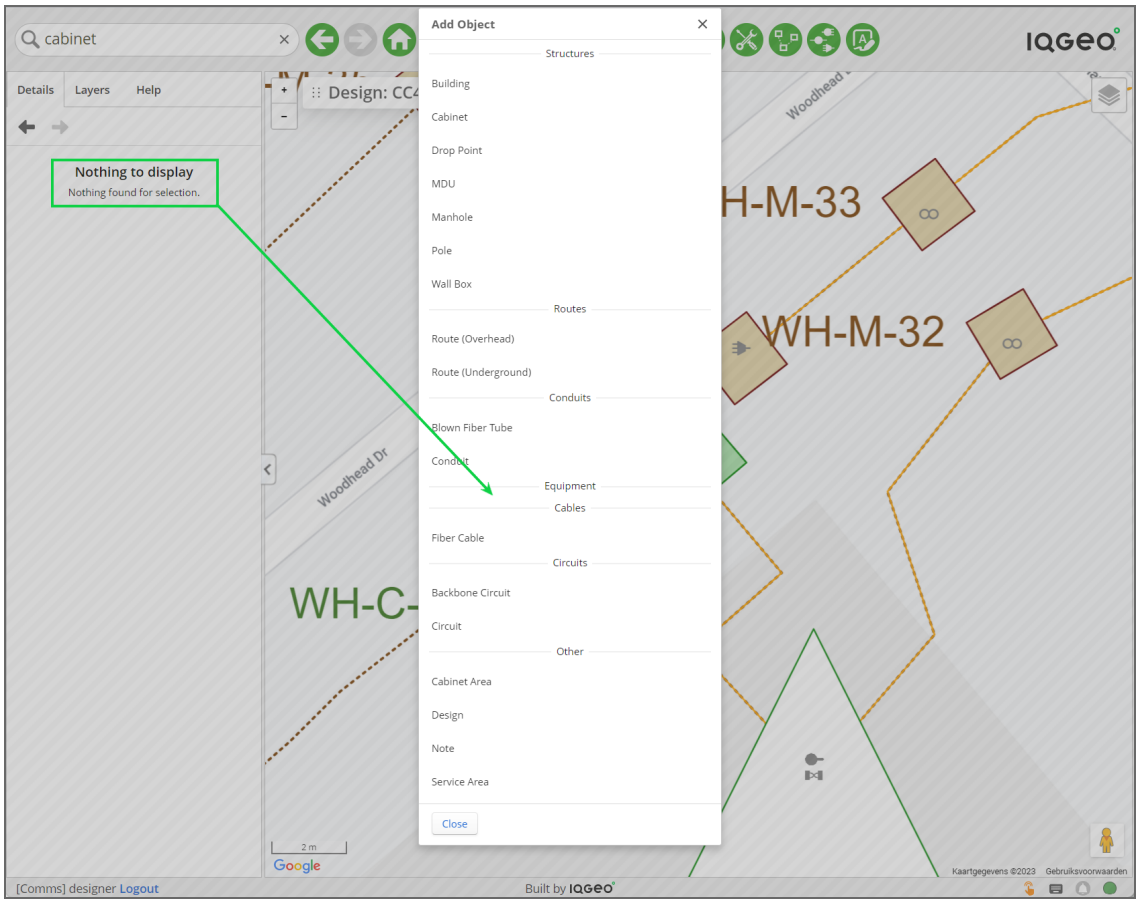

Figure 3-2: Add Object > Equipment - No Structure

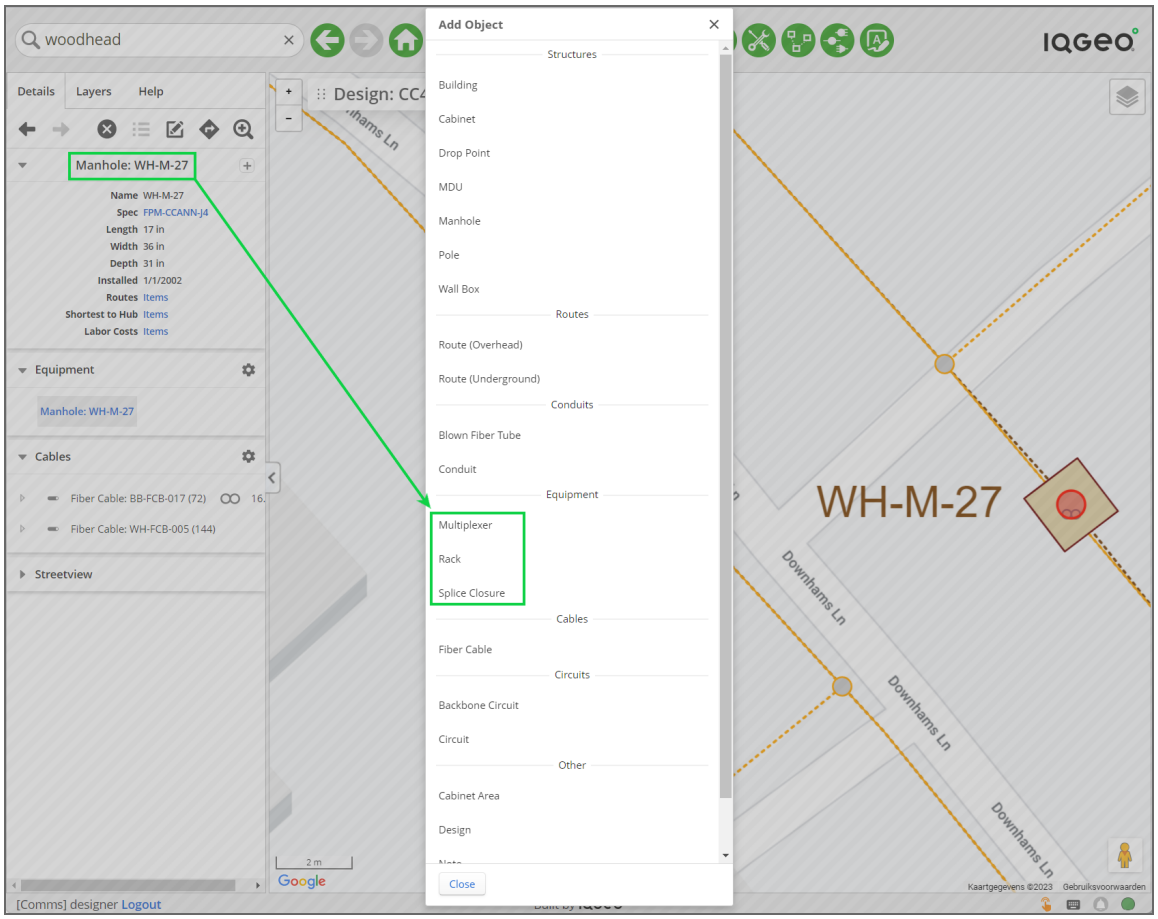

Figure 3-3: Add Object > Equipment - Manhole

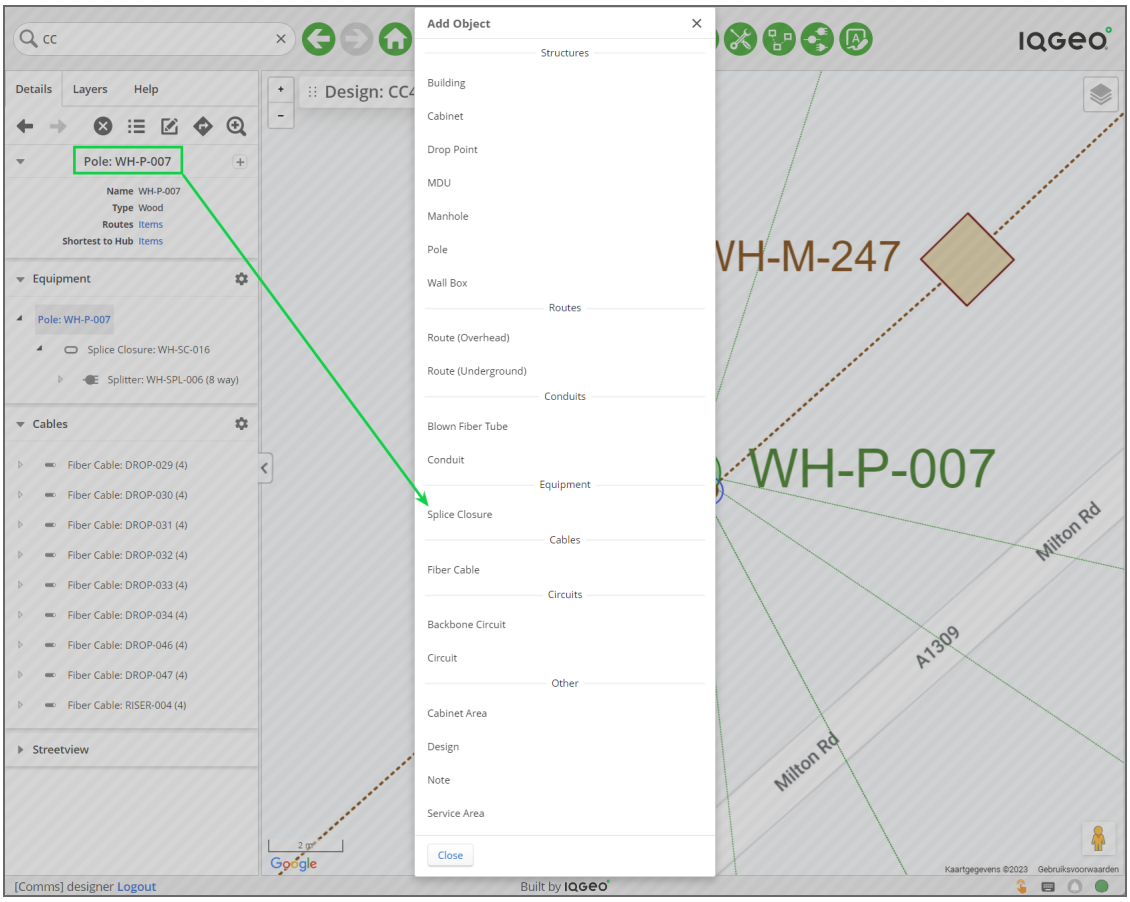

Figure 3-4: Add Object > Equipment_- Pole

3.2 Tools Palette

The following changes have been made to the tools in the **Tools Palette.**

3.2.1 Layout Strand

The Layout Strand feature provides an easy way to create overhead and / or underground routes. You simply click and drop points on the map to indicate which path the route should follow. Those points can be either plain vertices that determine the direction or they can be structures that have to be placed at certain locations on the route.

The route is automatically generated between the points you place.

1. Click the **Tools Palette** button on the toolbar at the top of the page. The associated tools appear on the right side of the map.

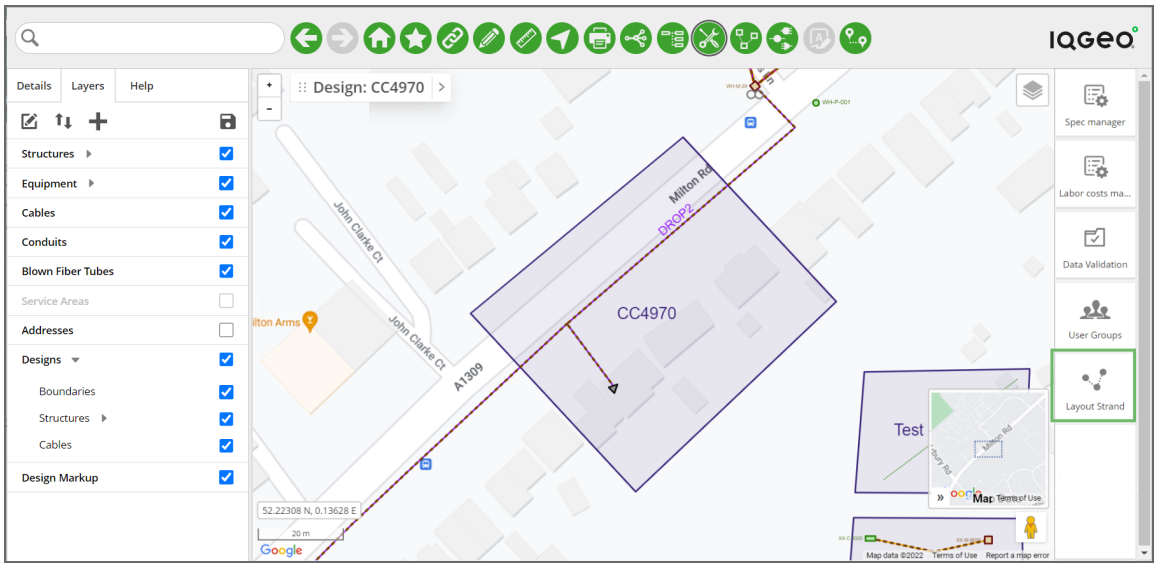

Figure 3-5: Tools Palette > Layout Strand Icon

2. Click the **Layout Strand** icon to display the **Strand Layout** dialog, then select the available features and routes for both overhead and underground.

> These are set in the Strand Layout settings configuration (see the Installation and Configuration Guide).

Overhead

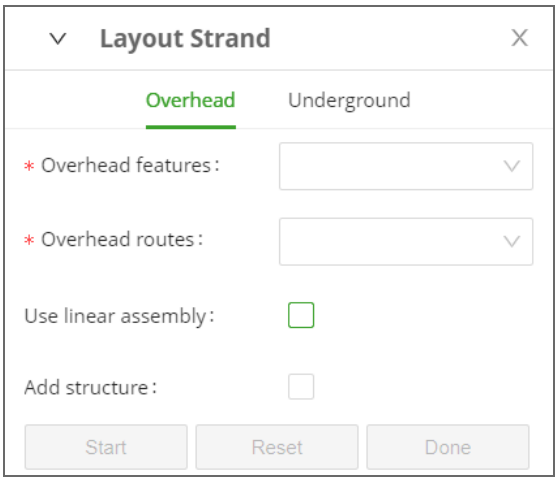

Figure 3-6: Layout Strand Dialog - Overhead Tab

3. Select or enter the following details:

- **Overhead features** - The physical features such as a *Pole* or *Building*.
- **Overhead routes** - The type of route the feature is to take.
- **Use linear assembly** - Enable the check box if you want to add access to any linear assemblies that have been created and then select the linear assembly required from the drop down list.

> The route is greyed out and cannot be selected if you are using a linear assembly.

- **Add Structure** - Enable this check box while drawing a route, every time you want to add a structure instead of a vertex on the map, at the location where you are about to click.

  This option is equivalent to holding the [`Ctrl`] key (Windows) or the [`Cmd`] key (Mac) while clicking a location on the map. It is primarily intended for users of the software on touch-screen devices, where the [`Ctrl`] key is absent.

> Fields that appear in the tool come from `mywcom.strandLayout` configuration. See the Installation and Configuration Guide.

Underground

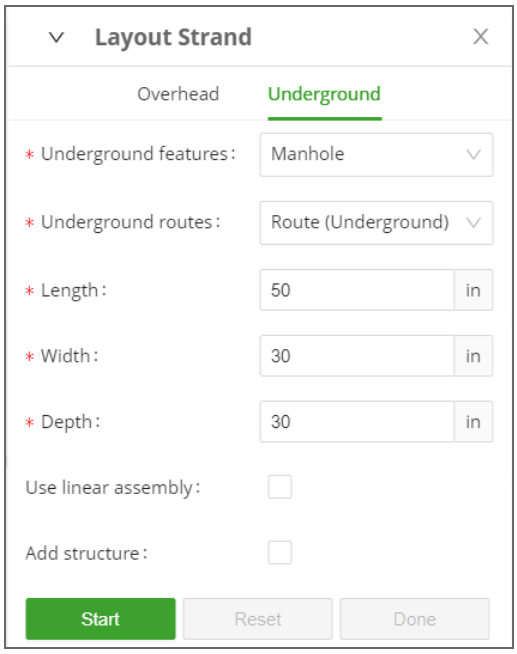

Figure 3-7: Layout Strand Dialog - Underground Tab

4. Select or enter the following details:

- **Underground features** - The physical features such as a *Manhole*.
- **Underground routes** - The type of route the feature is to take.
- **Use linear assembly** - Enable the check box if you want to add access to any linear assemblies that have been created and then select the linear assembly required from the drop down list.

> The route is greyed out and cannot be selected if you are using a linear assembly.

- **Add Structure** - Enable this check box while drawing a route, every time you want to add a structure instead of a vertex, at the location on the map where you are about to click. This option is equivalent to holding the [`Ctrl`] key (Windows) or the [`Cmd`] key (Mac) while clicking a location on the map. It is primarily intended for users of the software on touch-screen devices, where the [`Ctrl`] key is absent.

> Fields that appear in the tool come from `mywcom.strandLayout` configuration. See the Installation and Configuration Guide.

Once you have selected all the features / linear assemblies required, the **Start** button on the dialog is enabled.

The **Start** button is shown on both the **Overhead** and **Underground** tabs. When you click the **Start** button, you will be adding the feature / linear assembly shown on the corresponding tab.

5. Click the **Start** button and move the cursor to the position on the map where you want to start to place the feature / linear assembly.
6. Click the left mouse button to place the feature / linear assembly on the map.

> Fields and tab switching are all disabled while in “placing mode”.

7. Click the left mouse button to place the feature / linear assembly on the map.

- If you place a point that is a plain vertex of the path you are drawing, it is shown as a white square; if you place a point while the **Add** structure option is active, it is presented as a white square with a red circle around it.
- If you click a point of the path with the **Add** structure option active in a location where a structure already exists, no new structure will be placed on top of that. The route will connect to that existing structure.
- In the same way, if you move a point subsequently, that presents a structure, and place it on top of a point that already holds a structure, the structure of that former point will be taken out and vice versa.

> To ensure that this feature behaves as expected, it is important to place the points with necessary precision. If a newly placed point does not exactly coincide with the point where a structure is already present, a new structure can still be placed on top of the existing one.

You can click the **Reset** button on the dialog if you have placed the feature / linear assembly incorrectly. This enables you to place it in the correct location. You can also use the functions from the right-mouse context menu to make adjustments while you are drawing.

Linear assembly placing is indicated as shown in the following screen shot:

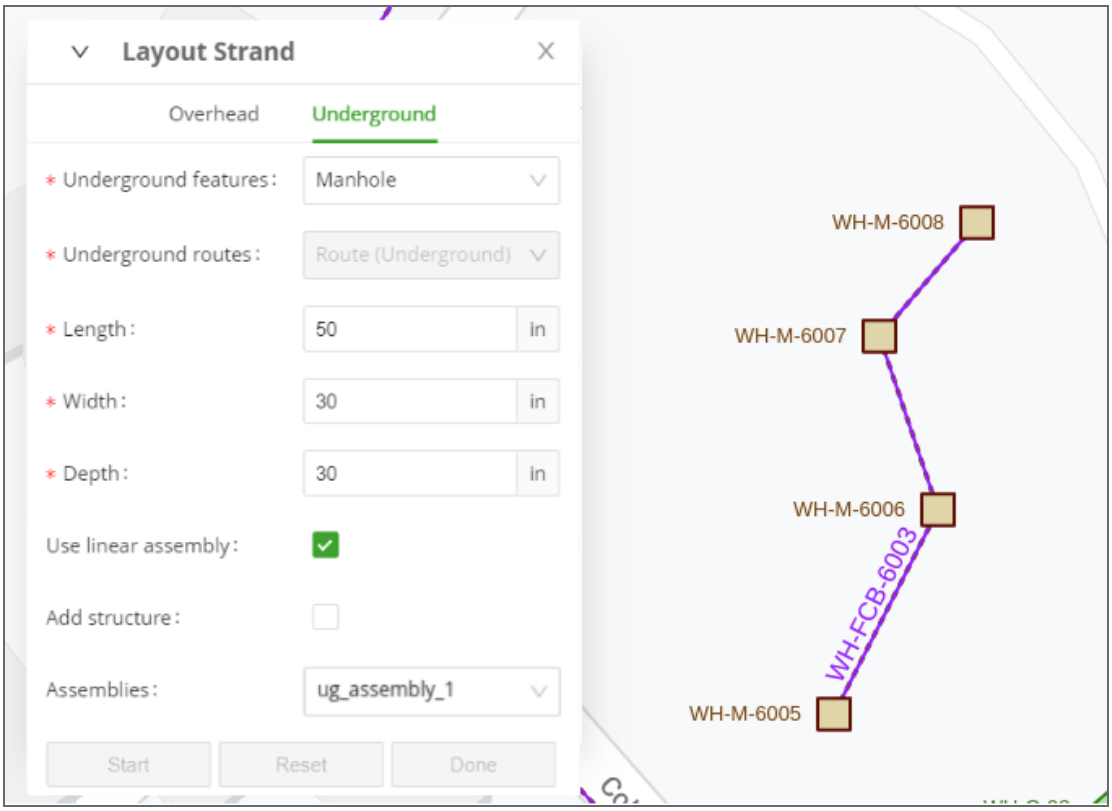

Figure 3-8: Linear Assembly Placing

8. Click the **Done** button on the dialog , once you have placed the feature / linear assembly in the required location. This adds the features, linear assemblies and routes to the database.

You can add further features /linear assemblies by clicking the **Start** button on the dialog again and placing it as described above.

3.2.2 Path Finder

The Path Finder feature allows you to search the available fiber paths between two structures you select in a design. You can view details of the found path(s) and then select one to create a circuit from it.

To activate the Path Finder feature, open a design.

> In the Designer role, this is required to obtain access to the Path Finder function.

Click the **Tools Palette** icon to open the tools shown on the right side of the map.

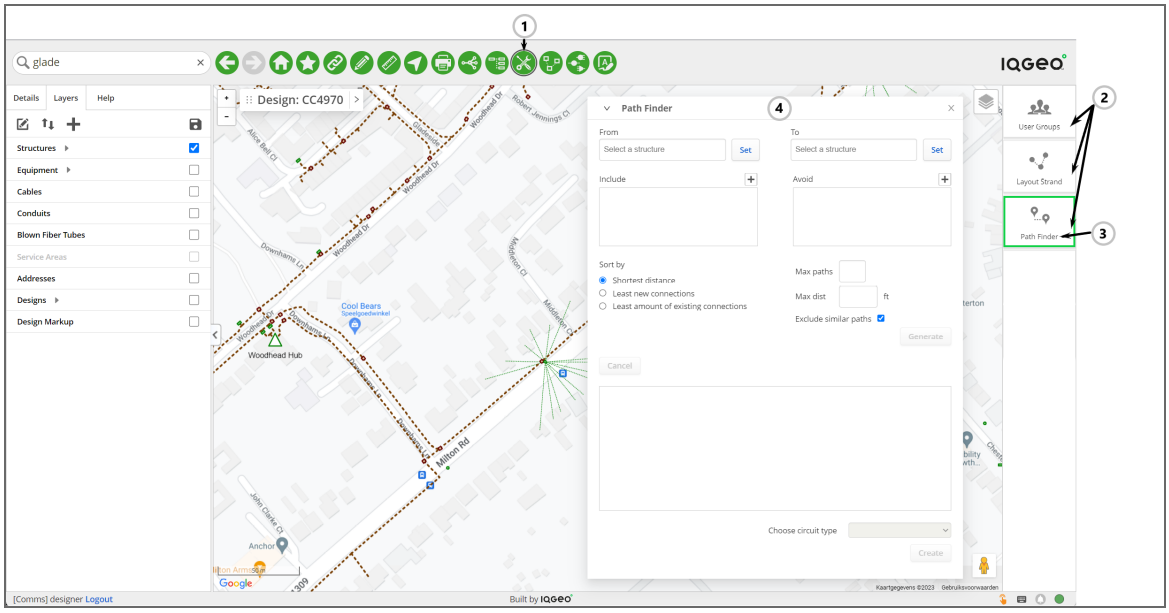

Figure 3-9: Tools Palette > Tools

1. **Tools Palette** icon
2. **Tool** buttons
3. **Path Finder** button.

Click the **Path Finder** icon. The **Path Finder** dialog is displayed. You can move the dialog window around in the Network Manager application window and collapse and expand it as required.

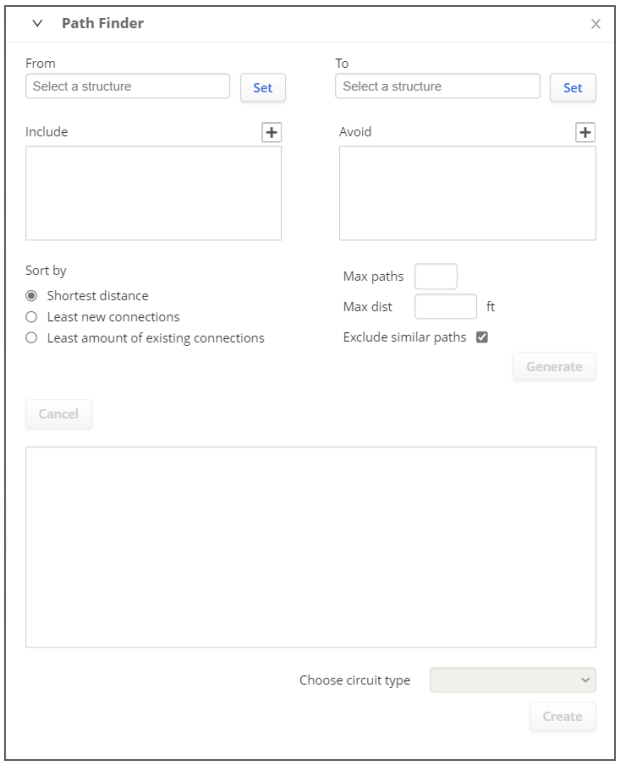

Figure 3-10: Path Finder Dialog

3.2.3 Performing a Path Finder Operation

Basic Settings

You must start by providing the basic settings for the operation.

Specify the structure to start from. You can do this directly on the map or use the **Search** tool to locate it, then click the **Set** button next to the **From** input field. The **Set Port** dialog appears.

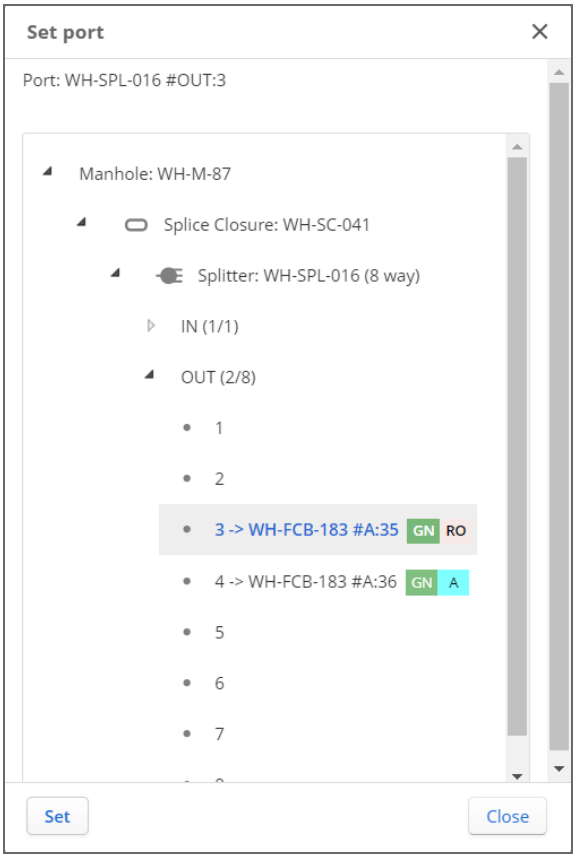

Figure 3-11: Set Port Dialog

You can expand the information tree up to the level of the port that you want to use as the starting point and click on it to select it and click the **Set** button.

The name of the selected port appears in the **From** input field.

Click the **Set** button next to the **To** input field and select the structure directly on the map or use the **Search** tool to locate it, then specify the target structure in the **To** field.

The name of the selected structure appears in the **To** input field.

> For the structure at the end location you do not have to set a specific port beforehand, but it is best to verify if ports are available to avoid that the operation terminates with the message that there is no port at the end location.

Additional Settings

You can provide additional settings, if appropriate, using the other fields in the **Path Finder** dialog.

Provide the names of certain elements in the design that *must be included* in the paths that you want to find and select a structure or a route or a piece of equipment and click the **+** button at the top right of the **Include** field.

Provide the names of specific elements in the design *that must be avoided* in the paths that you want to find and select a structure or a route or a piece of equipment. Click the **+** button at the top right of the **Avoid** field.

Supply the generic settings about the paths to be brought up, indicating:

- **Max paths** : The maximum number of paths you want to be produced.
- **Max dist** : The maximum length of the paths to be produced.
- Whether you want to **Exclude similar paths** : Select this option if you do not want multiple paths that overlap for most part.

Indicate how the results need to be sorted in the progress and results pane at the bottom of the dialog window.

You can select one of these sorting criteria:

- **Shortest distance** : If you prefer to see the shortest paths appear at the top of the results list.
- **Least new connections** : If you prefer to see the paths that require the least new connections at the top of the results list.
- **Least amount of existing connections** : If you prefer to see the paths that include the lowest number of existing connections at the top of the results list.

Click **Generate** to start the search for the available path(s).

The operation starts and progress information appears in the progress and results pane at the bottom. When Path Finder has completed its search, the result also appears in that pane.

> The direct feedback you obtain as a Designer from Path Finder allows you to monitor the operation that you have launched; if you need to monitor and / or manage the Path Finder task queue at a higher level (for example. as an Administrator), you can use the `comms_db` command-line Tool to do so. Refer to the `comms_db` Command-line topic in the Installation and Configuration Guide.

3.2.4 Assessing the outcome of a Path Finder operation

Path Finder does not necessarily produce a path or a list of paths. There can be different reasons for this. To determine if usable paths are available, Path Finder works in two phases:

- First, it looks for route paths between the start and the end location.
- And, subsequently, looks for fiber paths using those route paths.

With the data gathered during these two phases, it can construct and present the possible paths. Typical reasons why no paths can be produced are:

- *End location has no ports* : As mentioned above, no paths can be established if the structure at the end location does not have ports available.
- *No route paths found* : Path Finder's search for route paths did not yield (usable) results. This can also occur if you have set the **Max dist** parameter and route paths exceed this distance.
- *No paths found* : Path Finder was not able to detect any end to end path between start location and end location.

If Path Finder can generate paths between the two selected structures, it displays the result in the **Path Finder** dialog, as well as on the map.

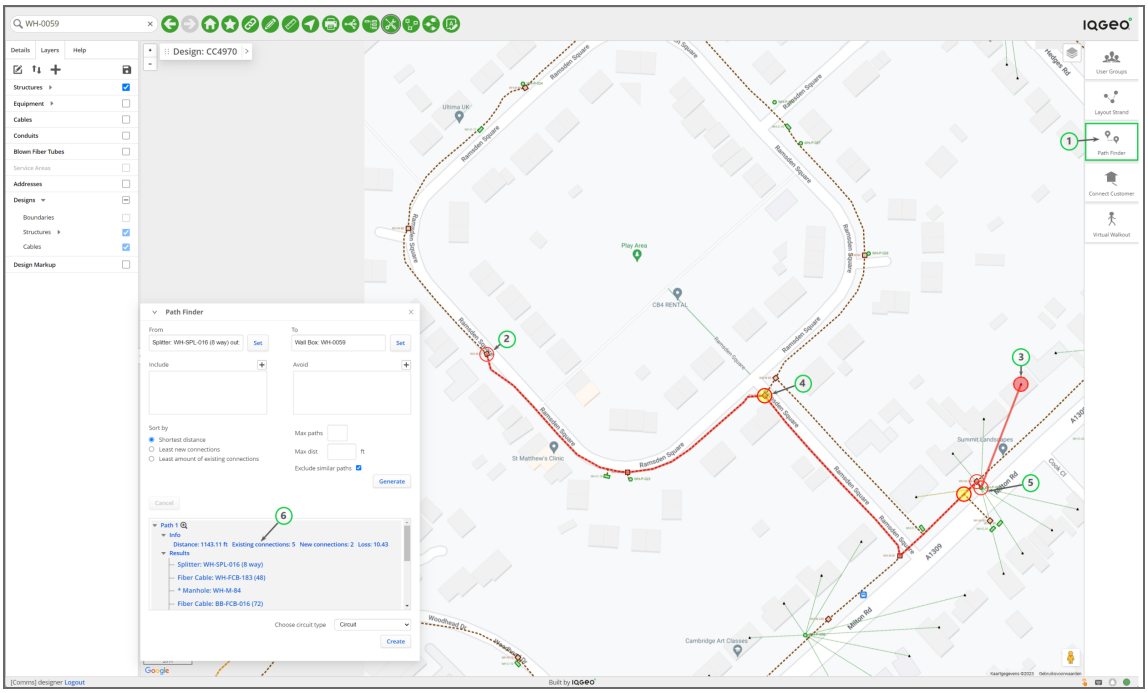

Figure 3-12: Path Finder Results displayed in Path Finder Dialog and on Map

1. Route generated by Path Finder
2. Structure at start location
3. Structure at end location
4. Existing connection
5. New connection to create
6. Detailed information on path(s) found.

In the results pane of the dialog, the information about path(s) is presented in a fixed format comprising these items:

- The identification of the path in the form "Path n" with 'n' a sequence number starting at 1 and incrementing for each subsequent path in the list.
- The **Zoom** icon, which allows to highlight the path on the map.
- The **Info**, which indicates the length of the path, the number of existing connections that are part of the path, the number of new connections that need to be created should this path be used to create a circuit and the optical loss occurring along this path.
- The **Results**, which consists of an overview of all the objects that are part of the path.

To assess the path(s) that Path Finder brought up, you can step through the information in the results pane, which interacts with the map. If you click the **Zoom** icon next to the path name, the map zooms to the path concerned. The selected path is displayed in red. All network objects that are part of the path are presented and highlighted on the map, and the connections that are part of it, are marked with a circle. Additionally, for new connections, the inside of that circle is highlighted in yellow.

If you hover an object in the **Results** overview, a pop-up window appears with a set of key characteristics of the object. At the same time, the object is highlighted on the map.

> The Path Finder function may provide a quick alternative for creating a circuit between the ports of two specific structures in your design.

3.2.5 To Create a Circuit along a Path

To create a circuit along a path:

1. Select the path that you want to use in the **Results** pane of the **Path Finder** dialog.
2. Choose the circuit type that you want to create from the list at the bottom of the dialog window.

> Circuit types are set in the Comms Settings configuration page (Administrators can refer to the Installation and Configuration Guide).

3. Click **Create** to generate the circuit.

The **Circuit** dialog appears.

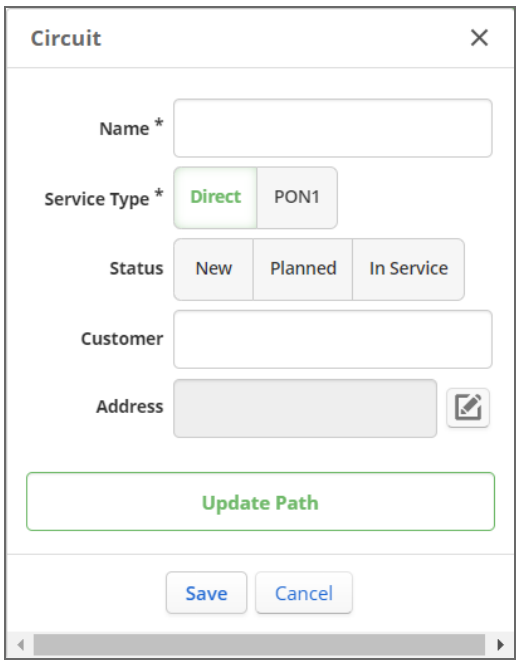

Figure 3-13: Circuit Dialog

4. Enter the details for the circuit you are creating:

The **Name** and the **Service Type** are mandatory fields.

5. Click **Save**.

If new connections are to be created, the **Insert Circuit** dialog appears, prompting for your approval to the creation of these new connections.

6. Click **OK** to confirm.

The circuit is created with all the equipment and connections required and is saved to the database.

4: Administrator GUI Changes

The following changes have been made to the Administrator GUI:

4.1 Configuration Pages

4.1.1 Settings - Advanced Tab

The following setting has been added to the **Advanced** tab of the **Settings** configuration page.:

Setting	Description	Example

mywcom.pathfinder	These settings are used to control the behavior of Path Finder and the creation of splice closures when it creates new connections for circuit routing.
	The `splice_closure_type` setting must be set to the name of the feature in the data model that will contain fiber connections, automatically created when routing a circuit along a path. The `splice_closure_properties` can be populated with default values for splice closure features created.
	For example:
	"splice_closure_type": "splice_closure",
	"splice_closure_properties": {
	"specification": "CS-FOSC400-A4-16-1-BNV"
	},
	The other settings can be modified to tune the Path Finder for particular networks.
	For details, contact IQGeo Support.
		{
		"max_nodes": 1000,
		"show_partial": false,
		"max_route_paths": 10,
		"max_route_nodes": 1000,
		"exclude_similar": false,
		"similarity_threshold": 90,
		"route_paths_only": false,
		"cable_paths_only": false,
		"bidirectional": false,
		"num_back_steps": 20,
		"max_paths_default": 5,
		"max_dist_default": 5000,
		"connection_cost_metres": 10,
		"splice_closure_type": "",
		"splice_closure_properties":
		{}, "include_as_via": true
		}

5: Data Model Changes

The following changes have been made to the Data Model:

1. The following network definitions have been added:
   - `mywcom_cable_segments` - Mesh network for tracing structures and fiber cable segments.
   - `mywcom_fiber_path` - Tree network for finding multiple candidate paths in fiber network.
2. Settings to support the new path finder functionality have been added. For details, see *Settings - Advanced Tab* on page 24
3. A new feature, `iqgapp_task_queue`, has been added to support running path finder requests asynchronously. The records for this feature can be managed using the `comms_db` manage_ tasks comment . See the *Command Line Tool* on page 26 section.
4. The change detail feature, `mywcom_change_detail`, has been made non-editable from the UI and can no longer be queried.

6: Tools

The following change has been made to the Tools in this release:

6.1 Command Line Tool

The *comms_db* Command

The *comms_db* command provides a number of comms-specific database operations. It can be used in combination with the IQGeo platform *myw_db* command to manage your database from the command line or within a script.

The format of the command is:

```
comms_db <db_name> <operation> ...
```

The following new operation is supported:

Operation	Description	
manage_tasks	Manage different aspects of tasks. The operation comprises a set of sub-operations for specific actions:	
	list	List tasks, their status and various details.
	cancel	Cancel a task with a given <task number>.
	delete	Delete all tasks that are not running or waiting for execution.
	run	Run a task with a given <task number>.

7: REST API

The following have been added to the REST API:

1. Path Finder
    a. Find fiber paths between two structures.
    b. Create circuit from specified path.
2. Asynchronous tasks
    a. Fetch status of task.
    b. Get log of task.
    c. Interrupt task.

8: Upgrading

This section provides an overview of the upgrade process for this release.

8.1 Upgrading Your Database

To upgrade from the previous release of Network Manager:

1. Apply all IQGeo Platform upgrades:

```
myw_db <db_name> upgrade core 650
```

2. Apply the Network Manager upgrade:

```
myw_db <db_name> upgrade comms 6526
```

3. Perform the post upgrade actions listed below.

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